

CareBot: Design of an Autonomous Follow-Me Medical Supply Carrier for Hospital Ward Rounds

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Abstract—Hospitals rely heavily on internal logistics systems to transport medical supplies, medications, and equipment between departments. In many cases, these tasks are still performed manually by medical staff using traditional carts. This can reduce the amount of time available for direct patient care and increase the physical workload of nurses and healthcare workers.

This project presents the conceptual design and engineering analysis of *CareBot*, a follow-me autonomous platform intended to assist hospital staff by transporting medical supplies inside hospitals and healthcare facilities. The robot is designed as a mobile cabinet capable of carrying up to 30 kg while safely navigating inside a hospital.

The development process follows the VDI 2221 systematic design methodology, which includes problem definition, conceptualization, design evaluation, and detailed engineering analysis. The proposed system integrates a differential drive platform powered by two DC motors, obstacle detection using time-of-flight sensors, and a modular battery system to support continuous operation.

I. INTRODUCTION

Hospitals depend on efficient internal logistics to make sure that medical supplies, medications, and diagnostic tools are available when needed. Many of these tasks are still done manually by nurses or hospital staff who transport equipment using traditional medical carts. Although these carts are simple and reliable, they require constant manual operation and can increase the workload of healthcare professionals.

Healthcare workers already face demanding work environments where time and physical effort are critical factors. Studies have shown that logistical tasks, such as transporting equipment and supplies, can occupy a significant portion of a nurse's working time [1]. With hospitals becoming more and more complex, and the patient demand keeps growing, improving internal logistics systems has become an important challenge.

Recent advances in robotics have made it possible to introduce autonomous mobile robots into hospital environments. These systems can assist with repetitive tasks such as supply delivery, allowing healthcare workers to focus more on patient care [2]. However, designing robots that can safely operate in hospitals presents many challenges. Hospital corridors are dynamic environments where patients, staff members, and medical equipment are in constant movement in common spaces.

The objective of this project is to design a mobile robotic platform capable of assisting hospital staff by transporting

medical supplies. *CareBot*, is designed as a follow-me robotic cart that can autonomously follow behind a user while carrying medical equipment. The platform is intended to operate safely in indoor hospital environments while carrying up to 30 kg of medical supplied.

The development of *CareBot* follows the VDI 2221 systematic design methodology, which provides a structured approach to engineering product development [3]. This methodology guides the design process from the problem definition and conceptualization, to the detailed engineering design and evaluation.

II. LITERATURE REVIEW

Robotic systems are being introduced more and more into healthcare environments to support hospital logistics and to help reduce the workload of medical staff. Hospitals require the frequent transportation of medications, medical equipment, and disposable supplies between departments and patient rooms. These tasks are often performed manually by healthcare professionals using traditional carts, which can consume valuable time and increase physical workload.

Autonomous mobile robots (AMRs) have been proposed as a solution to improve internal hospital logistics. Fragapane et al. explains that robotic transport systems can improve efficiency by automating repetitive material-handling tasks in healthcare facilities [2]. Several hospitals have already implemented robotic delivery platforms capable of transporting supplies between departments.

Examples of such systems include the *TUG* robot developed by Aethon, which is used in hospitals to transport medications, laboratory samples, and linens between different departments. Similarly, the *Moxi* robot developed by Diligent Robotics assists hospital staff by performing non-clinical tasks such as delivering equipment and supplies. These robots demonstrate the practical benefits of automation in hospital logistics.

However, most existing hospital robots are designed primarily for point-to-point delivery rather than directly assisting healthcare workers during routine activities. Research on assistive robotics suggests that robots capable of supporting hospital staff during daily tasks could improve workflow efficiency and reduce workload [4], [5].

Based on these observations, this project investigates the design of a follow-me robotic platform capable of accompanying healthcare professionals during ward rounds while

transporting medical supplies. This system could complement existing hospital delivery robots by directly assisting staff during everyday clinical activities.

III. CONCEPTUALIZATION PHASE

A. Problem Definition

Healthcare professionals are constantly transporting medical supplies such as medications, diagnostic tools, and disposable materials like gloves and masks, during routine hospital ward rounds. These supplies are typically carried using manual medical carts that have to be pushed through hospital corridors and patient rooms. Even though these carts provide organized storage, they require continuous manual handling and may limit the mobility of medical personnel.

Hospital environments are also dynamic spaces where patients, staff members, and medical equipment constantly move through corridors and treatment areas. Navigating these environments while pushing equipment can reduce efficiency and increase the physical workload associated with non-clinical tasks.

So problem we want to solve with this project, is the need for a more efficient and ergonomic way to transport medical supplies during hospital ward rounds. The proposed solution is an autonomous follow-me robotic platform capable of following healthcare workers while carrying essential medical supplies.

B. User Needs Analysis

The primary users of the proposed CareBot system are healthcare professionals such as nurses and doctors who perform daily ward rounds. These users need a system that can assist them with the transport of supplies without getting in the way of their clinical activities.

Several key user requirements were identified during the conceptual design stage:

- The system should operate with minimal user interaction.
- The robot must safely navigate crowded hospital environments.
- The robot should reliably follow a designated user during movement.
- The storage cabinet must provide organized and easily accessible compartments for medical supplies.
- The external surfaces should be easy to clean to maintain hospital hygiene standards.

C. System Requirements

Now, with the user needs identified, a set of requirements was defined for the CareBot platform. These requirements guide the design process and establish measurable performance targets.

TABLE I
PRELIMINARY SYSTEM REQUIREMENTS

Parameter	Target Value
Maximum system mass	50 kg
Payload capacity	20–30 kg
Maximum speed	0.8 m/s
Following distance	1 m
Robot width	≤ 600 mm
Turning capability	Differential drive (in-place rotation)
Obstacle detection range	≥ 2 m
Battery autonomy	≈ 4 hours

D. Design Constraints

Several constraints must be considered when designing a robotic system intended for hospital environments.

First, safety is a critical requirement because the robot will operate in close proximity to patients and healthcare staff. The system must therefore include reliable obstacle detection and emergency stop mechanisms.

Second, the robot must be compact enough to navigate hospital corridors and patient rooms without obstructing movement. This limits the overall dimensions of the platform.

Another constraint is battery capacity. The robot must be able to operate for several hours without requiring frequent charging in order to remain practical for hospital use.

Finally, the system must be built using commercially available components to ensure that the design remains technically and economically feasible.

E. Black Box Model

According to the VDI 2221 design methodology, the black box model is used to describe the system in terms of its main inputs and outputs without focusing yet on the internal technical solution [3]. In this project, CareBot is treated as a system that receives energy, user commands, and environmental information, and converts them into controlled movement and medical supply transport.

The main purpose of the system is to support hospital staff during ward rounds by carrying medical supplies while following the user safely through the hospital environment.

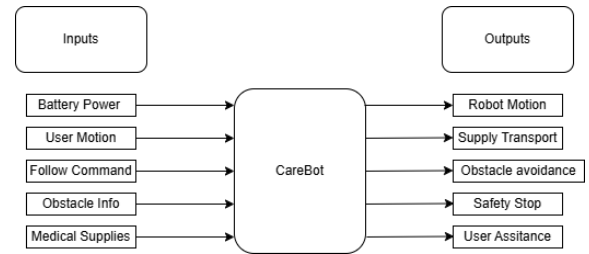


Fig. 1. Black box model of the CareBot system showing system inputs and outputs.

As shown in Fig. 1, the robot receives electrical energy from the battery, movement information from the user, and obstacle information from the surrounding environment. These inputs are processed by the system in order to generate

controlled movement, transport medical supplies, and maintain safe operation.

F. Functional Structure

After defining the black box model, the next step in the VDI 2221 approach is to break the overall function into smaller sub-functions [3]. This helps to understand what the system must do before choosing the final technical solution.

For CareBot, the overall function is to assist hospital staff by transporting medical supplies while autonomously following the user in a safe way. This overall function can be divided into several smaller functions, including storing supplies, providing power, detecting obstacles, processing user-following information, and generating wheel motion.

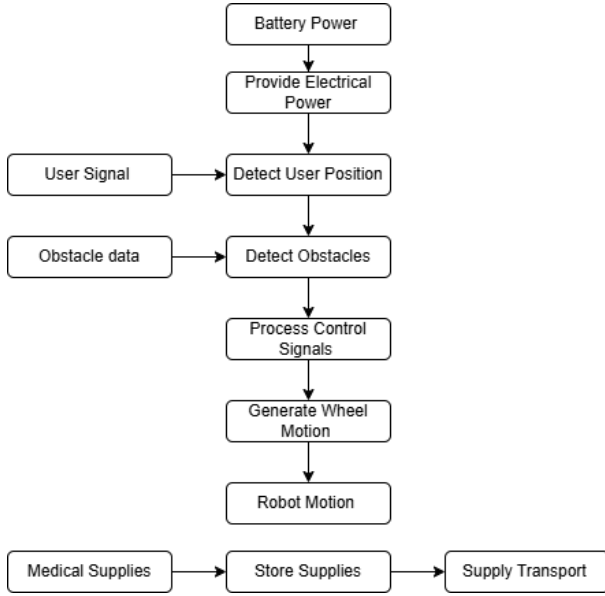


Fig. 2. Functional structure of the CareBot system.

The functional structure shown in Fig. 2 makes it easier to connect the user needs to the technical design of the robot. For example, the need for safe motion leads to the obstacle detection and control functions, while the need for supply transport leads to the storage and mobility functions. This functional breakdown also supports the next design step, which is the generation and comparison of different concept solutions.

G. Concept Generation

After defining the functional structure, the next step in the VDI 2221 design methodology is the generation of different technical solution concepts [3]. The purpose of this stage is to explore multiple design alternatives before selecting the most suitable architecture for the final system.

For the CareBot platform, three different mobility and control architectures were considered. Each concept satisfies the main functional requirements of transporting medical supplies and moving through hospital environments, but they differ in their drivetrain configuration, control architecture, and overall system complexity.

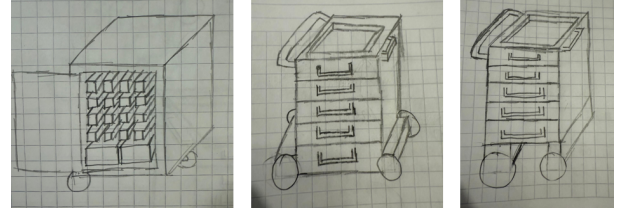


Fig. 3. Concept sketches generated during the conceptualization phase.

1) *Concept A: Three-Wheel Steering Platform:* The first concept uses a three-wheel configuration consisting of two drive wheels and a front steering wheel. In this architecture, the robot uses two BLDC motors for propulsion while a third motor controls the steering angle of the front wheel.

This design offers high maneuverability and smooth directional control. However, the addition of a steering motor increases mechanical complexity and requires a more advanced control system. The integration of multiple motor controllers also increases the overall system cost.

2) *Concept B: Differential Drive Platform:* The second concept uses a differential drive configuration with two powered wheels located on either side of the robot. The robot changes direction by varying the speed of the two motors, eliminating the need for a dedicated steering mechanism.

This architecture simplifies the mechanical design and reduces the number of moving components. The system can be controlled using a dual-channel motor driver and provides sufficient maneuverability for indoor environments such as hospital corridors. Because of its simplicity and reliability, differential drive systems are widely used in mobile robotics [9], [10].

3) *Concept C: Integrated Differential Drive System:* The third concept also uses a differential drive configuration but integrates a more advanced motor control system. In this case, high-performance BLDC motors with integrated encoders are controlled using a RoboClaw motor controller, which includes built-in feedback control capabilities.

This architecture offers high precision and improved motor control performance. However, the system is significantly more expensive and increases the overall electrical complexity of the robot.

The three proposed architectures were evaluated using a weighted decision matrix in order to determine the most suitable solution for the CareBot platform.

Subsystem	Option A	Option B (Selected)	Option C
Drive Motors	2× BLDC motors	2× Brushed DC gearmotors	2× BLDC motors
Steering	3rd motor with 360° rotating front wheel	Differential steering (speed difference)	Differential steering
Motor Controller	Dual BLDC ESC + steering driver	Cytron MDDS30	RoboClaw
Speed Feedback	Encoders	Magnetic Encoders (100-128 PPR)	Integrated encoders
Control Unit	Microcontroller	ESP32	RoboClaw internal control
Obstacle Sensor	None / Basic	ToF (VL53L1X)	Ultrasonic
Battery Type	Li-ion	Li-ion (7S, 35Ah)	LiFePO4
Total Motors		3	2
Estimated System Cost	High	Moderate	Very High
System Weight	Medium	Medium	High

Fig. 4. Comparison of proposed CareBot system architectures for the three design concepts.

The subsystem architecture of the three design concepts is summarized in Fig. 4.

TABLE II
WEIGHTED CONCEPT EVALUATION MATRIX

Criteria	Weight	Option A	Option B	Option C
Cost	0.30	2	5	1
Reliability	0.25	3	4	4
Complexity	0.15	2	4	3
Efficiency	0.15	4	3	4
Manufacturability	0.15	2	5	3
Total Score	1.00	13	21	15

TABLE III
ENGINEERING TRADE-OFF COMPARISON

Parameter	Option A	Option B	Option C
Mechanical Complexity	High (3 motors)	Low	Medium
Electrical Complexity	High	Medium	Low
Control Complexity	High	Medium	Low
Energy Efficiency	High	Moderate	High
Runtime Potential	Good	Very Good	Good
Expandability	Medium	High	Medium
Maintenance	Difficult	Simple	Moderate

IV. SYSTEM DESIGN

Following the concept evaluation stage, Concept B was selected to develop *CareBot*. The design phase focuses on translating the conceptual solution into a detailed system architecture that integrates mechanical, electrical, and control subsystems.

The CareBot platform is designed as an autonomous assistive robot capable of transporting medical supplies while autonomously following healthcare staff. The system is composed of several interacting subsystems including the mechanical structure, drive system, sensing system, power system, and control unit.

The mechanical structure consists of a mobile cabinet mounted on a differential drive base. Two driven wheels

provide propulsion and caster wheels ensure stability and smooth movement. The cabinet includes multiple drawers for organizing medical supplies and a flat upper surface that can be used as a temporary workspace during medical rounds.

The electrical and control architecture is centered around an ESP32 microcontroller, which is responsible for processing sensor data, controlling motor drivers, and executing the follow-me behavior. The drive motors are controlled using a Cytron MDDS30 dual-channel motor driver, allowing independent control of each wheel.

Obstacle detection and safety are ensured using multiple time-of-flight (ToF) distance sensors positioned around the robot [12]. These sensors allow the system to detect nearby obstacles and prevent collisions with patients, staff, or hospital equipment.

The system is powered by a removable lithium-ion battery pack operating at 24 V. This configuration allows the robot to operate for several hours while enabling quick battery replacement when necessary, which is particularly important in hospital environments where continuous operation may be required.

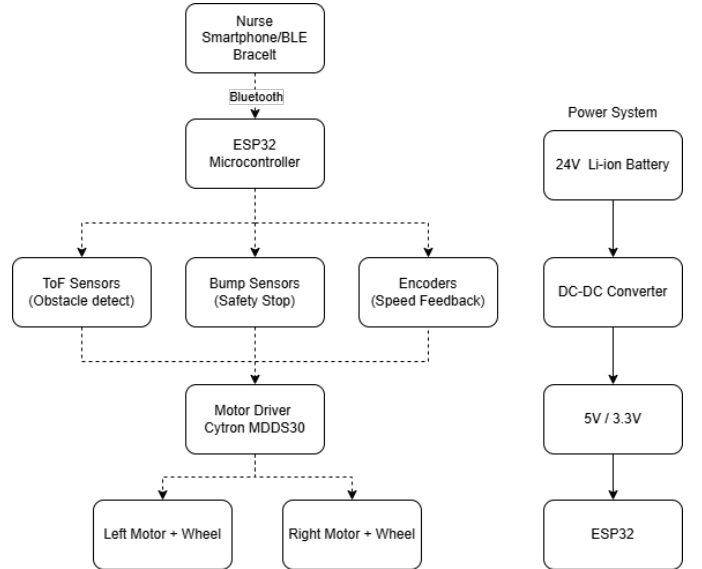


Fig. 5. High-level system architecture of the CareBot platform.

A. Mechanical Design

The mechanical structure of CareBot was designed to provide a stable mobile platform capable of safely transporting medical supplies within hospital environments. The design consists of a cabinet-style structure mounted on a differential drive mobile base.

The external cabinet dimensions were selected based on common hospital medical carts in order to maintain ergonomic usability for healthcare staff. The final dimensions of the platform are approximately 90 cm in length, 65 cm in width, and 120 cm in height. These proportions allow the robot to navigate through hospital corridors while maintaining sufficient storage capacity for medical supplies.

The cabinet structure includes multiple drawers designed for the organization of medical instruments, medications, and disposable supplies used during hospital rounds. A flat upper surface is provided to serve as a temporary workspace for healthcare staff when preparing materials or reviewing patient information.

The mobile base uses a differential drive configuration with two powered wheels located on each side of the chassis. This configuration allows the robot to rotate in place and maneuver efficiently in narrow hospital corridors. Two passive caster wheels are used in the front of the robot to ensure a smooth operation.

The structural frame of the robot is built mainly from aluminum profiles to provide enough strength while maintaining a relatively low system weight. The outer cabinet panels are manufactured using high-density polyethylene (HDPE), which is commonly used in medical environments due to its durability, chemical resistance, and ease of cleaning.

Based on the CAD model and estimated component masses, the total weight of the empty platform is approximately 20 kg. This allows the system to safely transport additional medical supplies while maintaining the design requirement of a total payload capacity of 50 kg.

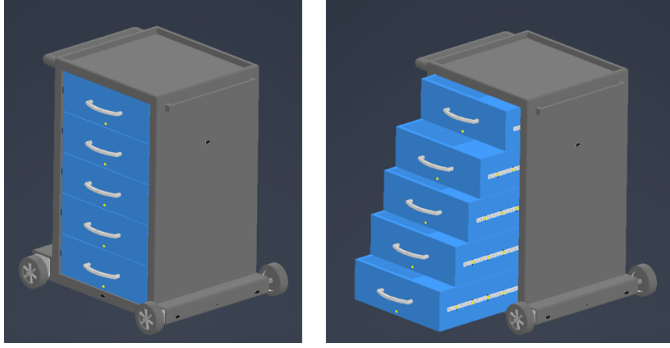


Fig. 6. 3D CAD model of the CareBot platform showing the cabinet structure and differential drive base.

B. Final Engineering Calculations

In order to justify the final design of CareBot, the main engineering calculations were carried out based on the selected system configuration. The robot was designed with a maximum total system mass of 50 kg, a wheel diameter of 150 mm, a maximum linear speed of 0.8 m/s, and a 24 V electrical system.

The main design assumptions used in the calculations are listed below:

- Maximum total system mass: $m = 50 \text{ kg}$
- Maximum linear speed: $v = 0.8 \text{ m/s}$
- Wheel diameter: $d = 0.15 \text{ m}$
- Wheel radius: $r = 0.075 \text{ m}$
- Number of driven wheels: 2
- Battery voltage: $V = 24 \text{ V}$
- Battery capacity: 20 Ah
- Ramp slope considered: 8.3%

1) *Wheel Circumference:* The wheel circumference was calculated from the selected wheel diameter:

$$C = \pi d \quad (1)$$

$$C = \pi \times 0.15 = 0.4712 \text{ m} \quad (2)$$

This means that one full wheel revolution moves the robot approximately 0.471 m.

2) *Required Wheel Speed:* The required wheel rotational speed was determined from the maximum target velocity:

$$n = \frac{v}{C} \quad (3)$$

$$n = \frac{0.8}{0.4712} = 1.698 \text{ rev/s} \quad (4)$$

This value was converted into revolutions per minute:

$$RPM = 60n \quad (5)$$

$$RPM = 60 \times 1.698 = 101.9 \approx 102 \text{ RPM} \quad (6)$$

Therefore, the wheel must rotate at approximately 102 RPM for the robot to reach a linear velocity of 0.8 m/s. This confirms the suitability of the selected 100 RPM gearmotor [8], [9].

3) *Gravitational Force:* The total gravitational force acting on the robot is

$$F_g = mg \quad (7)$$

$$F_g = 50 \times 9.81 = 490.5 \text{ N} \quad (8)$$

4) *Ramp Force:* To account for hospital ramps, a conservative slope of 8.3% was used. The force required to overcome the ramp is

$$F_{slope} = mg \sin \theta \quad (9)$$

Using $\sin \theta \approx 0.083$:

$$F_{slope} = 50 \times 9.81 \times 0.083 = 40.7 \text{ N} \quad (10)$$

5) *Rolling Resistance:* The rolling resistance force was estimated using a rolling resistance coefficient of $C_{rr} = 0.02$, which is consistent with values commonly used for wheeled vehicle analysis on smooth surfaces [11]:

$$F_{rolling} = C_{rr}mg \quad (11)$$

$$F_{rolling} = 0.02 \times 50 \times 9.81 = 9.81 \text{ N} \quad (12)$$

6) *Acceleration Force*: To include startup and moderate acceleration, the following relation was used:

$$F_{acc} = ma \quad (13)$$

Assuming an acceleration of $a = 0.25 \text{ m/s}^2$:

$$F_{acc} = 50 \times 0.25 = 12.5 \text{ N} \quad (14)$$

7) *Total Traction Force*: The total traction force required is the sum of the slope force, rolling resistance, and acceleration force:

$$F_{total} = F_{slope} + F_{rolling} + F_{acc} \quad (15)$$

$$F_{total} = 40.7 + 9.81 + 12.5 = 63.01 \text{ N} \quad (16)$$

8) *Required Wheel Torque*: The total torque required at the wheels is

$$T_{total} = F_{total} \cdot r \quad (17)$$

$$T_{total} = 63.01 \times 0.075 = 4.73 \text{ Nm} \quad (18)$$

Since the platform uses two driven wheels, the torque per wheel is

$$T_{wheel} = \frac{T_{total}}{2} \quad (19)$$

$$T_{wheel} = \frac{4.73}{2} = 2.37 \text{ Nm} \quad (20)$$

To provide a practical safety margin for turning, surface irregularities, and drivetrain losses, a motor capable of at least 4 Nm continuous torque per wheel was considered appropriate. The selected motor provides 5.63 Nm, which is above the calculated requirement.

9) *Angular Velocity*: The motor angular velocity was calculated from the selected wheel speed:

$$\omega = \frac{2\pi \text{RPM}}{60} \quad (21)$$

For 100 RPM:

$$\omega = \frac{2\pi \times 100}{60} = 10.47 \text{ rad/s} \quad (22)$$

10) *Mechanical Power per Wheel*: Mechanical power at the wheel was estimated using

$$P = T\omega \quad (23)$$

Using the minimum calculated torque requirement of 2.37 Nm:

$$P = 2.37 \times 10.47 = 24.8 \text{ W} \quad (24)$$

This is the theoretical minimum mechanical power required per wheel under the selected operating conditions. Since losses and dynamic effects must also be considered, the selected 90 W motor provides sufficient margin.

11) *Encoder Resolution*: Because the magnetic encoders are mounted directly on the wheels, the encoder resolution must be chosen based on wheel circumference [9].

The linear distance travelled per pulse is

$$\Delta x = \frac{C}{PPR} \quad (25)$$

For a 100 PPR encoder:

$$\Delta x = \frac{0.4712}{100} = 0.00471 \text{ m} \quad (26)$$

$$\Delta x \approx 4.71 \text{ mm/pulse} \quad (27)$$

For a 128 PPR encoder:

$$\Delta x = \frac{0.4712}{128} = 0.00368 \text{ m} \quad (28)$$

$$\Delta x \approx 3.68 \text{ mm/pulse} \quad (29)$$

Since the encoder is quadrature-based, the effective counts per revolution can be increased by 4x decoding:

$$CPR = 4 \times PPR \quad (30)$$

For 100 PPR:

$$CPR = 4 \times 100 = 400 \quad (31)$$

$$\Delta x = \frac{0.4712}{400} = 0.00118 \text{ m} \quad (32)$$

$$\Delta x \approx 1.18 \text{ mm/pulse} \quad (33)$$

For 128 PPR:

$$CPR = 4 \times 128 = 512 \quad (34)$$

$$\Delta x = \frac{0.4712}{512} = 0.00092 \text{ m} \quad (35)$$

$$\Delta x \approx 0.92 \text{ mm/pulse} \quad (36)$$

This resolution is more than sufficient for smooth speed regulation and basic odometry [8], [9]. For the final design, an encoder in the range of 100–128 PPR was considered appropriate.

12) *Battery Energy*: The selected battery is a 24 V, 20 Ah removable pack. The stored energy is calculated as [13]

$$E = V \cdot Ah \quad (37)$$

$$E = 24 \times 20 = 480 \text{ Wh} \quad (38)$$

13) *Power Consumption:* The two selected motors are rated at 90 W each, so the maximum theoretical motor power is

$$P_{motors,max} = 2 \times 90 = 180 \text{ W} \quad (39)$$

The low-voltage electronics, including the ESP32, sensors, and DC-DC converter losses, were estimated at approximately 20 W [13]. Therefore, the maximum total system power is

$$P_{total,max} = 180 + 20 = 200 \text{ W} \quad (40)$$

In practical operation, the motors are not expected to run continuously at rated load. A more realistic average system power was estimated as approximately 120 W.

14) *Battery Runtime:* The runtime was estimated using

$$t = \frac{E}{P} \quad (41)$$

Using the average operating power:

$$t = \frac{480}{120} = 4 \text{ h} \quad (42)$$

This confirms that the selected 24 V, 20 Ah battery can provide approximately four hours of practical operation [13].

15) *Average Battery Current:* The average current draw from the battery is

$$I = \frac{P}{V} \quad (43)$$

$$I = \frac{120}{24} = 5 \text{ A} \quad (44)$$

Under worst-case power demand:

$$I_{max} = \frac{200}{24} = 8.33 \text{ A} \quad (45)$$

This confirms that the selected 24 V system and motor driver are suitable for the expected operating conditions of CareBot.

C. Bill of Materials and Prototype Cost Estimate

A preliminary bill of materials (BOM) was prepared in order to estimate the prototype cost of the CareBot platform. The purpose of this table is to provide a realistic overview of the main hardware components required to construct a functional prototype.

For commercially available components, typical European retail prices were used. Mechanical parts such as the aluminum frame, HDPE panels, and mounting hardware were estimated based on small-scale prototype fabrication costs. Shipping, taxes, and manufacturing labor were not included in the total estimate.

Figure 7 illustrates the main mechanical components of the CareBot platform and their spatial arrangement within the system. The exploded CAD view helps visualize the integration of the cabinet structure, drive system, and electronic components used in the prototype.

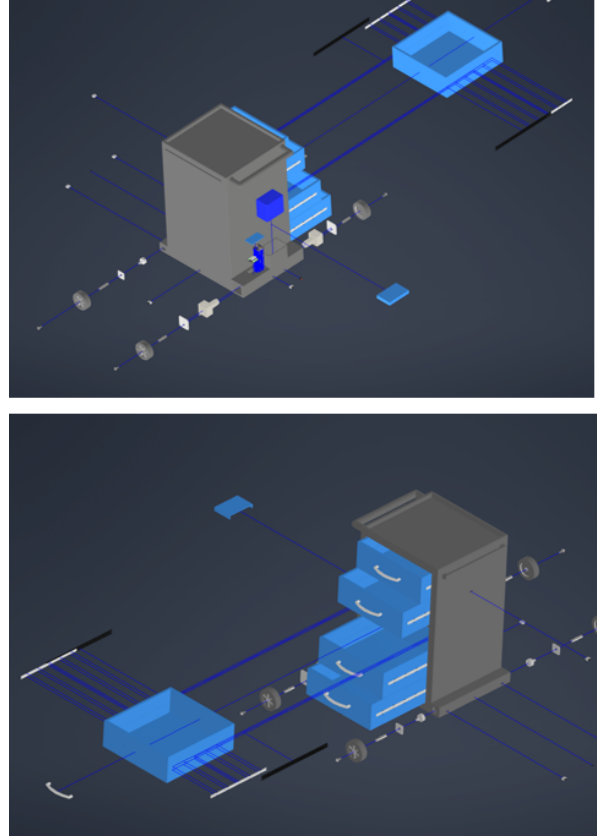


Fig. 7. Exploded view of the CareBot mechanical assembly showing the cabinet structure, drawer system, drive base, wheels, and internal electronics.

Table IV shows that the estimated prototype cost of the CareBot system is approximately 865 €. The most significant cost contributions come from the battery system, drive motors, and sensing components. These elements directly affect the robot's mobility, runtime, and safety capabilities.

It should be noted that the estimated cost corresponds to a single prototype unit. In a scaled production scenario, the cost per unit would likely decrease significantly due to bulk purchasing and optimized manufacturing processes.

TABLE IV
PROTOTYPE BILL OF MATERIALS FOR CAREBOT

Component	Qty.	Specification	Est. Cost (€)
DC gearmotor	2	24V, 90W, 100 RPM, 5.63 Nm (F5D90-24GU-30S/5GU30KB)	82
Motor driver	1	Cytron MDD530 dual motor driver	85
Microcontroller	1	ESP32 DevKit	8
ToF sensors	6	VL53L1X distance sensors	90
I ² C multiplexer	1	TCA9548A	7
Bump microswitches	3	Omron D2F series	3
Battery pack	1	24V 20Ah Li-ion battery	150
DC-DC converter	1	24V to 5V buck converter	20
Magnetic encoders	2	100–128 PPR wheel encoders	40
Drive wheels + casters	1 set	150 mm wheels + support casters	60
Aluminum frame material	1 set	6061 structural frame	120
HDPE cabinet panels	1 set	External cabinet panels	100
Telescopic drawer sliders	5 sets	Ball bearing drawer slides	20
Wiring, connectors, fuse, E-stop	1 set	Power and safety wiring	60
Bluetooth wearable tag	1	BLE tag or bracelet	20
Estimated Total			865€

D. Iteration and Design Optimization

During the design process several modifications were made to improve the performance, safety, and practicality of the CareBot platform. The initial concept was progressively refined as engineering calculations and feasibility considerations were carried out.

One of the first adjustments involved the physical dimensions of the platform. The initial concept proposed a wheelbase length of 120 cm. After evaluating the maneuverability in hospital corridors and considering the placement of the drive motors, the base length was reduced to approximately 90 cm. This modification improved turning capability and reduced the overall footprint of the robot.

Another design change was the obstacle detection system. The first concept used a LiDAR sensor for navigation. But, this approach required for the sensor to be mounted visibly on top of the robot, since it needs a 360° view. As a result, the sensing system was redesigned to use multiple Time-of-Flight (ToF) sensors positioned around the robot, combined with a front bumper safety sensor. This configuration provides reliable obstacle detection while maintaining a simpler system architecture.

The encoder resolution was also changed during the design process. Initially 48 PPR was estimated, but research showed that this resolution would not be good enough if the encoders

were mounted directly on the wheels. The final design therefore uses magnetic encoders with a resolution between 100 and 128 PPR, providing improved speed feedback and motion control accuracy.

Finally, the battery capacity and expected runtime were optimized. Even-though the idea was to have a very long battery life, up to 8 hours, the power analysis showed that a more realistic and practical configuration would use a removable 24 V, 20 Ah battery providing approximately four hours of operation. This solution allows batteries to be easily swapped during continuous hospital use and keeps the system weight down.

E. Risk Assessment

Since CareBot is intended to operate in hospital environments where the patients safety is critical, several potential risks were identified and evaluated during the design process. The risk assessment focuses on technical failures, operational hazards, and system reliability.

Table V summarizes the main risks associated with the CareBot platform, together with their potential consequences and mitigation strategies.

TABLE V
CAREBOT RISK ASSESSMENT

Risk	Possible Consequence	Likelihood	Mitigation Strategy
Obstacle detection failure	Collision with patients or equipment	Low	Multiple ToF sensors and bumper safety switch
Motor driver malfunction	Loss of mobility or uncontrolled motion	Low	Motor driver current limits and emergency stop
Battery overheating or failure	System shutdown or safety hazard	Low	Battery protection circuit and fuse protection
Communication loss (Bluetooth)	Robot stops following the user	Medium	Automatic stop when signal is lost
Wheel slippage on floor	Reduced motion accuracy	Medium	Closed-loop speed control using encoders
Sensor malfunction	Incorrect obstacle detection	Low	Redundant sensor placement and safety stop

The main safety concern for a mobile platform working in a hospital environment is the possibility of collision with patients, medical staff, or equipment. To eliminate this risk, CareBot uses multiple layers of safety. The primary obstacle detection system consists of several Time-of-Flight (ToF) sensors positioned around the robot. In addition, a front bumper equipped with mechanical microswitches acts as a final safety layer, ensuring that the robot stops immediately if a physical collision occurs.

Another potential risk is the loss of communication between the robot and the user device used for the follow-me functionality. In such cases, the control algorithm is designed to stop the robot automatically until a valid connection is restored.

Finally, electrical protection measures such as current limiting, fuses, and regulated power conversion are included in the design to reduce the risk of electrical faults. These safety considerations help ensure that the CareBot system can operate reliably within hospital environments.

F. Feasibility Study

After completing the conceptual design, the component selection, and the engineering calculations, a feasibility analysis was carried out in order to check if the proposed CareBot platform could realistically be designed and implemented. The feasibility of the system was evaluated from three different points: technical feasibility, economic feasibility, and manufacturing feasibility.

1) *Technical Feasibility*: From a technical point of view, the CareBot system is feasible using currently available components. The selected drive motors provide sufficient torque and speed to move the platform under the maximum design load of 50 kg. The torque calculations showed that each motor must provide at least 2.37 Nm of torque, and the selected motors are capable of delivering approximately 5.63 Nm. This provides a sufficient safety margin for ramp climbing and normal operation.

The sensing system is based on multiple Time-of-Flight (ToF) sensors distributed around the robot, providing reliable obstacle detection for indoor navigation. Combined with the bumper safety switch and wheel encoders, the sensing architecture allows the robot to operate safely within a hospital environment.

The ESP32 microcontroller provides sufficient computational capability to manage the sensor data, motor control, and Bluetooth communication simultaneously for the intended implementation at the prototype-level [9], [10]. In addition, the selected motor driver and power electronics are compatible with the 24 V electrical system used in the design.

2) *Economic Feasibility*: The estimated prototype cost of the CareBot system is approximately 865 €. Although this cost may appear relatively high for a single prototype, most of the cost is concentrated in reusable hardware components such as the battery, motors, and motor driver.

In a potential production scenario, the cost per unit could be significantly reduced through bulk component purchasing and optimized manufacturing processes. Therefore, from an economic perspective, the proposed system could be considered realistic for further development.

3) *Manufacturing Feasibility*: The mechanical structure of CareBot was designed using commonly available materials and manufacturing methods. The main structural elements use aluminum profiles, which are widely used in robotics prototypes due to their strength and ease of assembly.

The outer cabinet panels are made of HDPE, a material commonly used in medical environments because it is durable, resistant to chemicals, and easy to clean. These panels can be manufactured using standard CNC or laser cutting processes.

Since the majority of the system consists of commercially available components and simple mechanical parts, the con-

struction of a working prototype is considered feasible using standard equipment and fabrication tools.

V. WORKING PHASE

The working phase describes how the CareBot prototype would be implemented, integrated, and tested. Even-though the project was mainly focused on the design and engineering analysis of the platform, this section outlines the steps required to construct and evaluate a functional prototype.

A. Prototype Implementation

The construction of the CareBot prototype would begin with the assembly of the mechanical structure. The aluminum frame would first be assembled to create the base chassis of the robot. Then the 2 DC geared motors would be added, following the electronics and the wheels. This way we can test and debug any errors with ease of access to the components.

At this stage, the main objective would be to make sure that the robot can move correctly using the differential drive configuration. Basic motion tests would be performed to confirm forward movement, turning capability, motor control through the motor driver, and encoder feedback.

Once the drivetrain and electrical system are validated, the second stage of the prototype would integrate the sensing system. The Time-of-Flight sensors and bumper safety switches would be installed and connected to the ESP32 controller. Obstacle detection and safety stop functionality would then be tested.

After making sure that the platform works reliably, the final stage of the prototype would involve the installation of the cabinet structure. The HDPE panels, drawers, and telescopic drawer sliders would be mounted onto the aluminum frame to complete the external structure of the robot. This staged assembly approach reduces development risk and allows potential problems to be identified before the full system is constructed.

B. System Integration

After all components are installed, the hardware and software systems would be integrated. The ESP32 microcontroller would manage the communication between sensors, motor drivers, and the Bluetooth interface used for the follow-me functionality.

The motor driver would control the two DC gearmotors using pulse-width modulation (PWM) signals generated by the ESP32. Encoder feedback would allow the controller to regulate wheel speed and maintain stable motion [9].

The obstacle detection system would continuously monitor the environment using the ToF sensors. If an obstacle is detected within a predefined safety distance, the robot would automatically slow down or stop. The bumper switches provide an additional physical safety mechanism that immediately stops the robot in the event of contact.

Bluetooth communication would allow the robot to follow a user through a mobile device or wearable tag. The ESP32 would continuously estimate the relative position of the user and adjust the robot motion accordingly .

C. Testing and Validation

After the system integration, several tests would be performed to make sure that the CareBot platform works accordingly. Initial tests would focus on verifying the basic motion of the robot, including forward movement, turning, and stopping behavior.

Obstacle detection tests would then be carried out to ensure that the ToF sensors correctly identify objects at different distances. The bumper safety system would also be tested to confirm that the robot stops immediately when a collision occurs.

Additional tests would evaluate the robot's ability to follow a user while maintaining a safe distance. Finally, runtime tests would be performed to confirm that the selected 24 V, 20 Ah battery can provide approximately four hours of operation under typical conditions.

These tests would allow the performance and safety of the CareBot prototype to be verified before deployment in a real hospital environment.

D. Manufacturing Scale and Production Approach

For the CareBot platform, a medium-scale manufacturing approach is considered the most fitting production strategy. Hospital service robots are typically produced in relatively limited quantities compared to consumer products, since they are specialized systems designed for professional environments. Therefore, large-scale mass production methods such as fully automated manufacturing and extensive injection molding would not be necessary for the initial deployment of the system.

In a medium-scale production scenario, the mechanical structure of the robot could be manufactured using a combination of CNC machining, sheet metal fabrication, and modular aluminum frame components. The cabinet panels could continue to use HDPE or similar medical-grade plastics, which can be produced using CNC cutting or thermoforming processes. These manufacturing methods provide sufficient precision while allowing flexibility for design modifications.

Electronic components such as the motors, sensors, motor drivers, and microcontrollers are commercially available and can be assembled using modular wiring harnesses and standard connectors. For improved reliability in production units, the electronic components could be integrated into custom printed circuit boards (PCBs), reducing wiring complexity and simplifying assembly.

This production approach offers a balance between manufacturing efficiency and design flexibility. It allows the CareBot system to be produced in moderate quantities while maintaining the possibility of adapting the platform to the specific requirements of different hospital environments.

VI. CONCLUSION

This report presented the design and development of CareBot, a mobile follow-me robotic platform intended to assist healthcare personnel by transporting medical supplies within hospital environments. The objective of the project was to design a safe, practical, and technically feasible system capable of following a healthcare worker while carrying essential equipment during daily hospital operations.

The design process followed the VDI 2221 engineering methodology, beginning with the identification of the problem and the analysis of existing robotic solutions used in healthcare environments. Several conceptual designs were proposed and evaluated using a weighted decision matrix, which ultimately led to the selection of a differential drive mobile platform equipped with multiple obstacle detection sensors.

After selecting the final concept, the system architecture was developed to integrate the mechanical, electrical, and sensing subsystems. Engineering calculations were performed to determine key design parameters such as the required wheel speed, motor torque, encoder resolution, and battery capacity. These calculations confirmed that the selected components are capable of supporting the maximum system mass of 50 kg while maintaining a safe and reliable operation.

The final design includes two 24 V DC gearmotors providing 5.63 Nm of torque, a 24 V power system with a removable 20 Ah battery capable of approximately four hours of operation, and multiple Time-of-Flight sensors combined with bumper switches to ensure safe navigation in hospital environments. A bill of materials and prototype cost estimate were also developed, showing that the system can be constructed using commercially available components and standard manufacturing techniques.

The working phase of the project outlined the steps required to construct and test a prototype of the system, beginning with the development of the mobile chassis and drivetrain before integrating the sensing system and cabinet structure. In addition, a manufacturing and scalability analysis indicated that the system could be produced using medium-scale manufacturing methods suitable for specialized healthcare equipment.

Overall, the results of the design process indicate that the CareBot platform is technically feasible and has the potential to improve efficiency in hospital environments by assisting healthcare personnel with the transportation of medical supplies. Future work could focus on implementing more advanced navigation algorithms, improving the user-following system, and performing real-world testing within hospital facilities in order to further evaluate the performance and usability of the platform.

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